Performance Modeling of a Bank Customer Service System

1. Identify a Complex System

1.1 Selected System

The selected system for this performance modeling study is a bank customer service system. The system represents customers arriving at a bank branch, joining a waiting queue, receiving service from a customer service officer, and leaving after their transaction is completed.

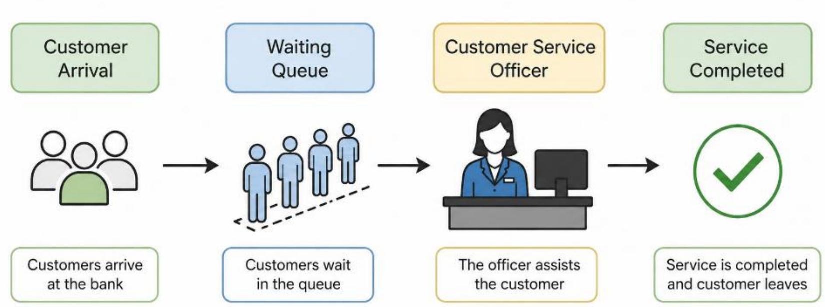
Although the basic process can be represented as a queue, the overall system has several performance factors that make it suitable for performance evaluation. Customer arrivals vary throughout the working day, different service periods can create different levels of congestion, and the available service capacity limits how many customers can be processed. These factors allow the system to be evaluated in terms of waiting time, queue length, throughput, resource utilization, bottlenecks, and scalability.

1.2 High Level Problem

Bank branches commonly experience variations in customer demand throughout the working day. During busy periods, a large number of customers may arrive within a short period of time. When customers arrive faster than they can be served, the waiting queue increases and customer waiting time becomes longer. This can negatively affect customer satisfaction and place greater pressure on service staff.

The main problem considered in this study is how to maintain an acceptable customer waiting time while making effective use of the available service capacity. The system must be able to handle normal customer demand while also responding to temporary increases in arrivals during peak periods. Therefore, performance modeling is required to understand how changes in customer arrival rates affect queue formation and waiting time.

1.3 System Process



In the proposed system, customers first arrive at the branch and join the waiting queue if the service officer is busy. When the officer becomes available, the next customer is served. After the transaction is completed, the customer leaves the system. This process provides measurable points for analyzing arrival rate, service rate, waiting time, queue length, and server utilization.

1.4 Why the System Is Suitable for Performance Modeling

Customer arrivals can be measured over fixed time intervals.

Service capacity can be represented using an estimated service rate.

Waiting time and queue length can be used as direct performance indicators.

Server utilization can show how effectively the available service resource is being used.

Peak arrival periods can be analyzed to identify potential bottlenecks.

Different staffing levels can be evaluated in future work to study system scalability.

2. Define Performance Objectives

The main performance objective of this study is to minimize customer waiting time while maintaining efficient use of the available service capacity. The following objectives will be considered during the performance evaluation.

Minimize customer waiting time: Measure and evaluate the average amount of time customers spend waiting before receiving service. Reducing this value is the primary performance goal.

Reduce queue length: Determine the average number of customers waiting for service and identify periods where the queue becomes congested.

Evaluate server utilization: Measure how much of the available service capacity is being used. A very high utilization may indicate that the system has limited capacity to handle sudden increases in demand.

Maintain adequate throughput: Evaluate the number of customers that can be served within a given period and determine whether the service capacity is sufficient for the observed demand.

Identify peak-period bottlenecks: Analyze variations in customer arrivals during the working day to identify periods where demand may exceed or approach available service capacity.

Evaluate scalability: In future analysis, compare different staffing configurations to determine whether adding another service officer can reduce waiting time and improve system performance.

3. Data Description and Methodology

Actual banking records were not available for this study. Therefore, a simulated dataset was created to represent realistic customer arrival patterns during a normal bank working day. Customer arrivals were recorded in 30-minute intervals from 9:00 to 17:00, giving a total observation period of eight hours.

The dataset contains the number of customers arriving during each 30-minute interval. The data is intended for performance modeling and will be used to estimate the average customer arrival rate for the queueing model.

Time Period	Customers Arrived
9:00–9:30	18
9:30–10:00	22
10:00–10:30	26
10:30–11:00	31
11:00–11:30	24
11:30–12:00	19
12:00–12:30	35
12:30–13:00	32
13:00–13:30	27
13:30–14:00	18
14:00–14:30	13
14:30–15:00	14
15:00–15:30	24
15:30–16:00	34
16:00–16:30	25

16:30–17:00	22
Total	**384**

Table 1: Simulated customer arrival dataset for the bank customer service system.

3.1 Dataset Summary

A total of 384 customer arrivals were recorded across the simulated eight-hour working day. The average arrival rate is calculated as follows:

λ = Total customer arrivals / Observation time

$\lambda = 384 / 8 = 48$ customers per hour

For the initial queueing analysis, the service rate is assumed to be 60 customers per hour, based on an average service time of approximately one minute per customer. This is a simplified assumption used for the initial M/M/1 model and can be refined using real service-time data in future work.

4. Initial Modeling Approach

The initial performance evaluation will use Queueing Theory, specifically an M/M/1 model, as a simplified first-order representation of the customer service process. The model assumes random customer arrivals, a single service resource, and exponentially distributed service times.

The main performance measures to be calculated in the next stage are server utilization, average queue length, average number of customers in the system, average waiting time in the queue, and average time spent in the system. Future work can extend the model to multiple service officers using an M/M/c model and can also consider more realistic service-time distributions.

5. Scope of the Current Submission

This document provides the high-level problem definition, identification of the selected complex system, performance objectives, and the simulated dataset required for the initial project submission. Detailed queueing calculations, visualizations, bottleneck analysis, and performance improvement evaluation will be carried out in the subsequent stages of the project.